

BASTION-LAB

Professional Scientific Assessment Report

Logminer: A Family-Aware Multi-Model Routing Architecture for Heterogeneous Log Anomaly Detection

Assessment of scientific value and recommendations for upgrading the manuscript to the quality level of a Class 1 international conference paper

Item	Information
Author of this report	Prof. Kya, Bastion-Lab
Destined to	First author
CC	All co-authors
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Assessment type	Internal pre-submission scientific review and improvement roadmap

Table 1. Administrative information for the assessment report.

This report evaluates the scientific value of the submitted Logminer manuscript and gives a concrete roadmap for lifting it from a promising applied prototype paper to a stronger international conference submission. The tone is intentionally direct: the manuscript has real potential, but a Class 1 conference paper requires a sharper scientific claim, stronger experimental evidence, better positioning against the state of the art, and a more reproducible artifact package.

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Note: page numbers may change slightly when the file is opened in another Word installation. The report was rendered and visually checked before delivery.

1. Executive verdict

Overall verdict: the manuscript has **good applied value** and a potentially useful architectural idea, but it is **not yet at Class 1 international conference level** in its present form. Its best current scientific asset is the idea of family-aware routing across heterogeneous log sources. Its most important weakness is that the paper currently demonstrates a useful prototype more than it proves a strong, generalizable scientific contribution.

The paper is scientifically meaningful because it addresses a real operational problem: security logs are heterogeneous, and forcing all sources through one monolithic detector can be technically fragile and analytically confusing. The paper also avoids a common weakness in cybersecurity ML papers: it does not blindly claim that every anomaly is an attack, and it admits that routing is controlled specialization rather than a universal F1-score improvement.

However, a Class 1 conference paper needs more than a reasonable architecture and several dataset results. It needs a clearly isolated contribution, strong baselines, a defensible experimental protocol, statistical robustness, reproducibility, strong related work, and a convincing explanation of why the proposed method advances the state of the art rather than only integrating known components.

Dimension	Current assessment	Reason
Scientific value	Promising but not yet mature	The problem is relevant and the architecture is practical, but the scientific contribution needs sharper proof.
Novelty	Moderate	The family-aware routing idea is useful, but it must be distinguished clearly from mixture-of-experts, ensemble routing, SIEM decoders, and modular IDS architectures.
Experiments	Useful first evidence	The current experiments show feasibility, but not enough causal evidence or statistical robustness for a top-tier venue.
Writing	Reasonably clear	Claims are cautious, which is good, but the manuscript must become more compact, more focused, and more argumentative.
Current likely level	Workshop / good regional or mid-tier conference	With revision, it can become a solid international paper; with major experiments, it can approach Class 1.
Class 1 path	Major revision required	Add factorial ablation, stronger baselines, artifact release, statistical evaluation, and clearer novelty framing.

Table 2. Executive quality verdict on the submitted Logminer manuscript.

2. Current scientific value of the manuscript

The current manuscript is valuable because it identifies a practical gap between ordinary log management and heterogeneous AI-based anomaly detection. The paper correctly observes that Windows logs, Linux/authentication logs, Wazuh exports, HDFS, BGL, CICIDS2017 and UNSW/CIC-DDoS do not share the same structure, semantics or feature contracts. This observation is not trivial for real security operations: a detector that works well on one family can be misleading on another.

The scientific value is currently strongest at the **system architecture** level. The paper proposes a pipeline with collectors, parsers, normalization, routing, detection, correlation, dashboarding and audit. It also gives an auditable routing score and a fallback mechanism. These are useful engineering choices for a lightweight security analytics prototype.

The scientific value is weaker at the **algorithmic and experimental proof** level. The paper explicitly states that it does not introduce a new detector. That is acceptable only if the architectural contribution is proven rigorously. At the moment, the experimental evidence is useful but not yet sufficient to prove that the architecture is fundamentally better than strong alternatives.

Aspect	Assessment	Detailed comment
Problem relevance	High	Heterogeneous log anomaly detection is a real challenge in SOC/SIEM environments.
Architecture design	Good	The modular decomposition is understandable and practically useful.
Formalization	Medium	The routing equations help, but the scoring policy is rule-based and needs calibration or sensitivity proof.
Experimental breadth	Good but uneven	Many sources are included, but some are treated as candidates rather than validated labels.
Experimental depth	Insufficient for Class 1	The paper needs stronger baselines, repeated runs, confidence intervals and factorial ablations.
Reproducibility	Incomplete	The paper should release code, configurations, seeds, dataset preprocessing scripts and trained model cards.
Impact potential	Good	A deployable lightweight gateway architecture could interest applied cybersecurity communities.

Table 3. Scientific value assessment by dimension.

3. Strengths that should be preserved

The manuscript contains several strengths that should not be lost during revision. The target should not be to rewrite the paper into a generic deep-learning log anomaly paper. The interesting part is precisely the family-aware architecture and the careful claim that the method is a complement to existing tools, not a replacement for mature SIEM/HIDS platforms.

- The paper targets an operationally meaningful problem: heterogeneous log streams with different formats and semantics.
- The manuscript uses cautious language and avoids claiming that unsupervised anomalies are confirmed intrusions.
- The architecture is modular and understandable: collection, parsing, normalization, routing, detection, correlation and audit each have a defined role.
- The routing formula makes the family decision auditable. This is important for security analysts because a black-box route would be difficult to trust.
- The paper includes a controlled common-space ablation and an operational specialization ablation, which is stronger than merely reporting one final F1-score.
- The paper reports important negative evidence, especially the weak HDFS result under row-level features. This honesty improves credibility.
- The dashboard and correlation layer show that the work is not only a classifier experiment but an integrated workflow prototype.

These strengths point toward a good paper, but they must be supported by stronger evidence and cleaner framing. The main revision strategy should be: keep the architecture, sharpen the claim, expand the experiments, and make the evidence reproducible.

4. Main weaknesses blocking Class 1 acceptance

The current manuscript would face several serious reviewer objections at a Class 1 conference. These objections are not fatal, but they must be addressed before submission.

Item	Reviewer concern	Correction before submission
Contribution too architectural	The paper says it is not a new detector; reviewers may ask what the core research novelty is.	Define Logminer as a calibrated family-aware routing and specialization framework with measurable benefits and failure modes.

Item	Reviewer concern	Correction before submission
Baselines too limited	Comparisons against standard tools and modern log anomaly models are not enough.	Add DeepLog/LogAnomaly/LogBERT-style baselines for sequence logs, tabular baselines for network data, and modular baselines for routing.
Ablation incomplete	The paper admits that a full factorial ablation is future work.	Move the factorial ablation into the paper, not future work.
No statistical robustness	Single F1 values are not enough for top-tier review.	Report repeated runs, confidence intervals, variance across seeds and hyperparameter sensitivity.
HDFS weakness unresolved	The method fails badly on HDFS false positives under current features.	Add a sequence-aware HDFS/BGL branch or present HDFS as a limitation with a strong corrective experiment.
Related work too short	Eleven references are not enough for a mature international paper.	Expand to 35-50 references covering log parsing, log anomaly detection, IDS, SIEM, mixture-of-experts and artifact evaluation.
Operational claims under-proven	Local FastAPI tests do not prove SOC-scale feasibility.	Add a streaming throughput and backlog experiment with realistic ingestion rates and failure recovery.

Table 4. Critical weaknesses and required corrections for Class 1 positioning.

5. Detailed assessment by scientific dimension

The following scorecard uses a strict scale. The point is not to discourage the authors; the point is to identify where the manuscript must be strengthened to compete with high-quality international papers. A score above 8/10 means nearly Class 1 ready. A score between 6 and 8 means promising but needs revision. A score below 6 means major additional work is required.

Criterion	Current	Target	Gap
Problem importance	8	9	The problem is relevant, but the paper should connect it more clearly to SOC workload and heterogeneous enterprise monitoring.
Novelty	6	8	Routing is useful, but the relation to mixture-of-experts, ensemble routing and SIEM decoder architectures must be clarified.
Technical rigor	6	8.5	Routing formalization exists, but weight calibration, sensitivity and fallback validity are weak.
Experimental design	6	9	Datasets are broad, but protocols, baselines, splits and statistical tests must be stronger.
Result interpretation	7	9	The cautious discussion is good, but the paper must convert negative results into a stronger scientific narrative.
Reproducibility	5	9	Need code release, exact scripts, model cards, seed policy and environment files.
Writing quality	7	9	Readable, but too much narrative. Needs sharper thesis, shorter claims and stronger figures.
Class 1 readiness	5.5	8.5	Major revision and additional experiments are required.

Table 5. Class 1 readiness scorecard.

The current paper should not be submitted unchanged to a top-tier venue. It should first go through a controlled strengthening phase. If the team can add the experiments and references recommended in this report, the paper can become a strong international submission.

6. Contribution and novelty: how to sharpen the paper

The paper currently says that the main contribution is not a new detector but a family-aware multi-model routing strategy. This is honest, but it also creates a risk: reviewers may interpret the work as an engineering integration of existing pieces. The paper must therefore explain why the architecture is scientifically non-trivial.

The contribution should be reframed around three ideas: **auditable routing**, **controlled specialization**, and **evidence-based fallback**. The router is not just an if-else script if it is evaluated as a measurable scientific component: it assigns family confidence, selects compatible feature/model branches, detects unsupported inputs, and exposes the route reason to analysts. That can be a strong contribution if it is proven rigorously.

Current phrase	Problem	Recommended stronger formulation
The main contribution is not a new detector.	Sounds like the paper has no algorithmic novelty.	The contribution is an auditable family-aware specialization layer that controls detector selection under heterogeneous log semantics.
Routing should be interpreted as controlled specialization.	Good but abstract.	Routing is evaluated as a mechanism that enables compatible feature spaces, calibrated fallback and family-specific detection policies.
The system is a lightweight software-agent architecture.	Reviewer may expect real autonomous agents.	Use the term modular software-agent workflow, and explicitly state what autonomy is and is not present.
Logminer complements SIEM/HIDS tools.	Good but needs proof.	Position Logminer as an analytical model-selection layer that can sit above or beside Wazuh/OSSEC/ELK exports.

Table 6. Recommended reframing of the contribution.

A stronger title may also help. The current title is descriptive. A sharper title could be: **Logminer: Auditable Family-Aware Routing for Heterogeneous Log Anomaly Detection**. This title makes the scientific promise clearer: not merely multi-model, but auditable routing under heterogeneity.

7. Methodology and experimental design: what must be strengthened

The methodology is understandable, but it must be made more falsifiable. The key question should be: **When does family-aware routing improve detection quality, analyst usability, robustness or operational control compared with monolithic alternatives?** This question requires experiments that separate routing, feature space and detector choice.

The current paper has a controlled common-space ablation and an operational specialization ablation. This is a good start. However, the paper itself admits that a full factorial ablation is future work. For a Class 1 venue, that experiment should be part of the submitted paper. Without it, reviewers can say that the improvement comes from feature engineering or model choice, not from routing.

Experimental axis	Configurations	Purpose
Router	Global / rule-based family-aware / learned or calibrated router	Does routing itself help, and under which confidence conditions?
Feature space	Common minimal / family-native / sequence-template / hybrid	Is the gain due to routing or to better family-specific features?
Detector	Same detector family / different specialized models / sequence-aware models	Is the gain due to model specialization?
Correlation	Off / simple time-host grouping / weighted incident correlation	Does incident grouping reduce analyst workload without hiding errors?
Fallback	No fallback / fallback only / calibrated unknown-family fallback	Does fallback improve robustness to unknown or corrupted logs?

Table 7. Minimum factorial ablation needed for a stronger scientific claim.

The revised paper should include at least one table that explicitly states which hypotheses are supported, partially supported or rejected. The paper should not aim to win every dataset. A high-quality paper can be accepted if it explains why the method works in some families and fails in others.

8. Datasets, baselines and metrics: minimum evidence package

The manuscript uses several relevant sources: Windows logs, Wazuh exports, Linux/auth logs, HDFS, BGL, CICIDS2017 and UNSW/CIC-DDoS. This breadth is a strength, but the datasets are not equivalent. Some are supervised, some are unsupervised, some are sequence-oriented, and some are network-flow datasets. A Class 1 paper must not put them all into one undifferentiated basket.

The revised manuscript should group datasets by data family and by evaluation validity. For labeled datasets, use F1, precision, recall, false positive rate, PR-AUC, ROC-AUC where meaningful, and confusion matrices. For unlabeled operational logs, report anomaly candidate rates, analyst triage criteria and qualitative incident examples, not detection accuracy. For sequence logs such as HDFS and BGL, include sequence-aware baselines.

Dataset family	Current use	Recommended improvement
Linux/auth	Random Forest with F1 and FPR	Add baseline logistic regression, XGBoost/LightGBM if available, calibrated thresholds, seed variance and error analysis.
CICIDS2017	Strong Random Forest F1	Avoid saturated scores; report class-wise results, imbalance handling and attack-family breakdown.
UNSW/CIC-DDoS	Very high F1	Verify split leakage risk; use strict temporal or chunk-based split with no duplicate leakage.
HDFS	Weak row-level result	Add Drain + sequence model, DeepLog-style baseline, LogAnomaly/LogBERT-style comparison if feasible.
BGL	Strong result	Report sequence and template-based results, not only row-level model performance.
Windows/Wazuh	Anomaly candidates	Add analyst-labeled sample or synthetic attack injection to estimate precision.
Unknown/corrupted	Robustness check	Turn into a stress-test suite with clear fallback precision, rejection and failure-cost metrics.

Table 8. Dataset-specific upgrades required before Class 1 submission.

9. Results and interpretation: how to avoid overclaiming

The current manuscript has a valuable quality: it avoids claiming universal superiority. This should be preserved. In fact, the controlled common-space ablation shows that the global model can match or even slightly outperform family-aware models on some datasets. This is not a problem if the paper frames the result correctly.

The final claim should be: **family-aware routing is not automatically better than a global model; it is useful when it enables compatible feature spaces, meaningful fallback, interpretable routing reasons and family-specific operational policies.** That is a more mature and defensible claim than saying that routing always improves F1.

Observation	Bad claim	Good claim
CICIDS nearly saturated	The architecture is universally superior.	CICIDS is easy or saturated under the chosen features; improvement is not meaningful there.
Linux/auth improves in operational setting	Routing alone caused the gain.	Family-aware routing plus native features and specialized model improved performance.
UNSW improves slightly	Large scientific breakthrough.	Small gain; report confidence intervals and class-wise effects.
HDFS false positives high	Method failed globally.	Row-level lightweight features are insufficient for sequence logs; route HDFS to sequence-aware branch.

Observation	Bad claim	Good claim
Fallback preserves corrupted records	Robust detection of unknown threats.	Graceful degradation and traceability, not confirmed attack detection.

Table 9. Safer interpretation of the current results.

The results section should include at least three error-analysis examples. For instance: one false positive from Linux/auth, one HDFS false positive generated by the missing sequence context, and one fallback case where the system correctly preserved an unknown record. Class 1 papers often win reviewer trust by showing failures clearly.

10. Reproducibility, artifact and engineering quality

For a Class 1 conference, especially in applied cybersecurity and machine learning, reproducibility is no longer optional. The paper must make it possible for reviewers or later readers to understand exactly how the results were obtained. If the code cannot be fully released, a restricted artifact, detailed pseudo-code and exact configuration package are still necessary.

Component	Required content
Repository structure	Clear folders for data preprocessing, parsers, routing, detectors, correlation, dashboard, experiments and figures.
Environment	requirements.txt or environment.yml, Python version, operating system and hardware.
Data scripts	Download or preprocessing scripts for public datasets; clear policy for non-public Windows/Wazuh samples.
Model cards	For each detector: family, features, training set, test set, metrics, limitations and intended use.
Router policy file	A readable configuration file containing all routing weights, thresholds and fallback rules.
Random seeds	Fixed seed policy and repeated-run protocol; report variance.
Figure/table scripts	Scripts that regenerate all figures and tables from result files.
Audit examples	Sample JSONL traces showing route decision, detector output, correlation and analyst-facing incident.

Table 10. Minimum reproducibility package for a strong international submission.

The current paper already mentions local JSONL traces, audit and workflow logs. These should be turned into an explicit artifact contribution. The authors should include one appendix figure or table showing a full event journey: raw log -> normalized event -> family route -> model output -> anomaly candidate -> incident -> dashboard view.

11. Writing, structure, figures and tables

The manuscript is generally understandable, but it must be made more compact and visually stronger. Several tables and figures are dense. Figures with small text will be difficult to read in conference format. The paper should reduce visual clutter and make each figure answer one question.

Paper element	Problem	Correction
Abstract	Clear but long and cautious	Make the claim sharper: problem, method, evidence, limitation in 180-220 words.
Introduction	Good motivation, but contribution could be sharper	Add one paragraph that states exactly why monolithic detection fails and what Logminer proves.
Related work	Too short	Expand and classify literature into log parsing, log anomaly detection, IDS/SIEM, routing/mixture-of-experts and evaluation validity.
Figures	Several figures are too small	Use fewer, larger figures. Replace some bar charts with summary tables if space is tight.

Paper element	Problem	Correction
Tables	Informative but dense	Keep only the tables that support a claim; move implementation details to appendix or artifact.
Limitations	Good and honest	Move from defensive tone to scientific insight: limitations explain the regime map.

Table 11. Writing and presentation improvements.

A recommended structure is to make the first two pages extremely strong: introduce the heterogeneity problem, show one motivating example, state the architecture, and preview the ablation results. The reader should understand by the end of page two what is new, why it matters and how it is tested.

12. Section-by-section revision instructions

Part of paper	What is good	What to improve
Title	Family-aware and heterogeneous log wording	Consider emphasizing auditable routing and controlled specialization.
Abstract	Cautious claims	Add precise best evidence: which families improve, where it does not improve, and why.
Introduction	Problem motivation	Add one strong example showing why a global detector can confuse families.
Related work	Foundational IDS and log parsing references	Add modern log anomaly detection, sequence models, transformer-based log models, SIEM evaluation and mixture-of-experts/routing work.
Methodology	Agent roles and normalized contract	Add pseudo-code for routing and fallback; add model-card style table.
Routing mechanism	Equations and confidence score	Add sensitivity analysis and justification of routing weights.
Implementation	FastAPI/JSONL/dashboard explanation	Separate scientific method from implementation details; move operational internals to artifact section.
Experiments	Dataset breadth	Add full factorial ablation, stronger baselines, repeated runs, class-wise metrics and runtime scalability.
Results	Careful discussion	Add confidence intervals, statistical tests and error cases.
Limitations	Honesty	Convert limitations into a regime map and future experiment plan.
Conclusion	Controlled specialization	End with exact scientific takeaway, not only future work.

Table 12. Section-by-section revision instructions for the first author.

13. Suggested new paper architecture

A stronger version of the paper should be structured around hypotheses and evidence. The current organization is conventional, but a Class 1 version should be more argumentative. The paper should lead the reviewer from problem to claim to evidence to limitations.

Proposed paper section	Role in the revised manuscript
1. Introduction	Define heterogeneous log anomaly detection and state the controlled-specialization claim.
2. Background and related work	Separate log parsing, anomaly detection, SIEM tools, routing/ensembles and evaluation pitfalls.
3. Logminer architecture	Explain modules, normalized contract and event journey.
4. Auditable family-aware routing	Formalize score, confidence, fallback and calibration/sensitivity.

Proposed paper section	Role in the revised manuscript
5. Experimental protocol	Define hypotheses, datasets, splits, baselines, metrics and artifact policy.
6. Results	Report controlled ablation, operational ablation, sequence-log branch, robustness, runtime and error analysis.
7. Discussion	Explain where routing helps, where it does not, and how Logminer complements existing tools.
8. Threats to validity	Present internal, external, operational and reproducibility threats.
9. Conclusion	State the final scientific takeaway and artifact availability.

Table 13. Proposed structure for a stronger conference submission.

The authors should avoid trying to present every implementation detail in the main paper. The main paper must prove the scientific contribution. Implementation details, router constants, dataset manifests and screenshots can be placed in an appendix or artifact repository.

14. Concrete experiments to add before submission

The following experiments are the most important practical steps. They are ordered by expected scientific impact. If time is limited, prioritize E1 to E5. Without these experiments, a Class 1 submission will remain fragile.

ID	Experiment	Scientific purpose	Deliverable
E1	Full factorial ablation	Separates routing, feature space, model choice and correlation.	Table with main effects and interactions.
E2	Sequence-aware HDFS/BGL branch	Fixes the largest scientific weakness in current results.	Drain-like parser + sequence detector baseline comparison.
E3	Repeated-run statistical evaluation	Shows robustness beyond one split/seed.	Mean, standard deviation, confidence interval, significance test.
E4	Router sensitivity and calibration	Proves that routing constants are not arbitrary.	Sensitivity plot/table for weights, thresholds and fallback rate.
E5	Unknown-family stress test	Tests graceful degradation under unsupported formats.	Fallback precision, rejection rate, analyst triage examples.
E6	Streaming and backlog experiment	Tests whether local quasi-real-time claims are credible.	Throughput, latency percentiles, memory, backlog under burst.
E7	Analyst incident-quality evaluation	Tests whether correlation helps real human workflow.	Small labeled incident set or expert review of candidates.
E8	Comparison to Wazuh/OSSEC/ELK export workflow	Clarifies complementarity rather than replacement.	Case study showing Logminer adds model routing and correlation.

Table 14. Priority experiments required for a serious Class 1 upgrade.

The most critical experiment is E1. It converts the paper from a prototype description into a causal scientific evaluation. The second most critical is E2 because the current HDFS result will otherwise be attacked by reviewers familiar with sequence-based log anomaly detection.

15. Reference and related-work expansion plan

The reference list is currently too short for a mature conference paper. Eleven references can be acceptable for a short workshop paper, but not for a strong international submission on log anomaly detection. The revised paper should include approximately 35 to 50 carefully selected references, not a random long bibliography.

Reference area	Purpose	Concrete coverage
Classical IDS and anomaly detection	Gives foundation and taxonomy.	Denning, Axelsson, Chandola, Sommer and Paxson, Buczak and Guven.

Reference area	Purpose	Concrete coverage
Log parsing	Necessary because parser quality affects detection.	Drain, Spell, AEL, template mining and parsing benchmarks.
Log sequence anomaly detection	Necessary for HDFS/BGL.	DeepLog, LogAnomaly, LogRobust, transformer/log-BERT style models.
Network intrusion datasets	Necessary for CICIDS/UNSW validity.	Dataset papers, critiques of leakage and imbalance, modern evaluation protocols.
SIEM/HIDS systems	Clarifies complementarity.	Wazuh/OSSEC, ELK/SIEM log management, NIST log management guidance.
Routing and ensemble specialization	Clarifies novelty.	Mixture-of-experts, model selection, domain routing and OOD/fallback detection.
Evaluation and reproducibility	Strengthens credibility.	Artifact evaluation, ML reproducibility, calibration and confidence intervals.

Table 15. Related-work expansion plan.

The first author should not add references mechanically. Each new reference must serve a role: define a baseline, explain a limitation, position the contribution, or justify an evaluation choice. The related work should end with a gap statement: existing tools either parse, detect, visualize or manage logs, but few papers rigorously evaluate auditable model routing across heterogeneous log families.

16. Action plan and responsibility matrix

The revision should be managed as a mini-project. The paper will not reach Class 1 level through language polishing alone. It requires additional experiments, stronger references, code organization and a disciplined rewrite.

Task	Lead	Timing	Deliverable
Rewrite contribution and abstract	First author + Prof. Kya	Week 1	New title, abstract, contribution bullets.
Expand related work	Two co-authors	Week 1-2	35-50 references with classified literature map.
Implement factorial ablation	Technical lead + ML co-author	Week 2-4	Ablation table and statistical analysis.
Add HDFS/BGL sequence branch	Log parsing lead	Week 3-5	Sequence-aware baseline and comparison.
Reproducibility package	Implementation co-author	Week 4-6	Repository, configs, model cards, scripts.
Rewrite results and limitations	First author	Week 5-6	Controlled claims and regime map.
Internal co-author review	At least two co-authors before Prof. Kya	Week 6	Written validation and comments.
Final Prof. Kya review	Prof. Kya	After co-author validation	Final scientific decision.

Table 16. Practical revision plan for the first author and co-authors.

The previous internal instruction remains important: before the draft reaches Prof. Kya again, at least two co-authors should have read it critically, improved it and given a positive opinion. This avoids sending a premature draft to the senior reviewer.

17. Submission positioning and target venues

The right venue depends on how far the authors can improve the paper. If the paper remains mainly a prototype with limited new experiments, it should target a good applied cybersecurity, software engineering or systems workshop. If the authors complete the factorial ablation, sequence-log branch, statistical evaluation and artifact release, the paper can target stronger international conferences in cybersecurity analytics, dependable systems, applied AI for security or data mining applications.

Level	Expected manuscript quality	Best submission strategy
Level A: current version	Architecture plus initial prototype results	Workshop, regional/international applied conference, demo track.
Level B: moderate revision	Improved writing, better references, some extra experiments	Good specialized international conference or symposium.
Level C: major revision	Full ablation, strong baselines, sequence branch, artifact, statistics	Class 1-oriented submission in security analytics, applied ML systems or data mining.

Table 17. Submission strategy according to revision maturity.

The authors should not rush submission if the target is truly Class 1. A weak early submission can lead to predictable rejection and lost time. A disciplined additional experimental cycle is more valuable than cosmetic polishing.

18. Final recommendation to the first author

The Logminer paper has a real core idea: heterogeneous logs should not be treated as one homogeneous data stream, and auditable family-aware routing can be useful when it enables compatible detectors and interpretable incident candidates. This is a worthwhile idea. The current manuscript already contains honesty, modularity and useful first results. These are good foundations.

However, the paper must become more scientific and less prototype-descriptive. The first author should now focus on proof. The key question is not whether Logminer can run; the key question is whether Logminer teaches the community something reliable about family-aware specialization for heterogeneous log anomaly detection.

Recommendation: do not submit the manuscript immediately to a Class 1 venue. First, carry out a major revision with stronger related work, full factorial ablation, sequence-aware HDFS/BGL experiments, repeated-run statistics, calibrated routing analysis and a reproducible artifact package. After that, the paper can become a serious international conference submission.

The expected attitude is not to defend the current draft at all costs. The expected attitude is to improve it until the contribution is obvious to a skeptical reviewer. A good Class 1 paper should make the reviewer say: this problem is real, this architecture is justified, the evidence is rigorous, the limitations are honest, and the artifact can be inspected.

Appendix A. Suggested rewritten contribution statement

Suggested contribution statement for the revised paper:

1. We introduce Logminer, an auditable family-aware routing architecture for heterogeneous log anomaly detection. It normalizes diverse log sources into a common event contract while preserving family-specific evidence needed for model selection.
2. We formalize a routing and fallback mechanism that assigns log batches to compatible detector branches with explicit confidence and route explanations, allowing unsupported or corrupted sources to degrade gracefully rather than silently fail.
3. We evaluate controlled specialization across system logs, SIEM exports and network-flow datasets using common-space and operational ablations, false positive analysis, workflow latency and robustness checks.
4. We show that family-aware routing is not a universal F1 improvement, but a controlled specialization mechanism that is most valuable when heterogeneous families require different feature contracts, detector branches and analyst-facing explanations.

Appendix B. Suggested abstract skeleton

Security monitoring pipelines often receive heterogeneous logs whose syntax, semantics and feature spaces differ across operating systems, SIEM exports, distributed systems and network flows. Applying one monolithic detector to all sources can hide family-specific structure and create poorly interpretable alerts. This paper introduces Logminer, an auditable family-aware routing architecture for heterogeneous log anomaly detection. Logminer normalizes events into a common contract, assigns each batch to a log family with explicit route evidence, selects a compatible lightweight detector and

correlates anomaly candidates into analyst-readable incidents. We evaluate the architecture across Windows, Wazuh, Linux/auth, HDFS, BGL, CICIDS2017 and UNSW/CIC-DDoS data. Controlled and operational ablations show that routing should be interpreted as controlled specialization rather than a universal F1 improvement. Results indicate that the approach is useful when family-specific feature spaces and detector policies are required, while sequence-oriented logs such as HDFS require dedicated parsing and temporal detection branches. Logminer is positioned as an analytical complement to mature SIEM/HIDS platforms, not as a replacement.

Appendix C. Final pre-submission checklist

Item	Pass criterion
Contribution	One-sentence novelty is clear and defensible.
Related work	At least 35-50 relevant references, organized by theme.
Baselines	Strong baselines for each family, including sequence baselines for HDFS/BGL.
Ablation	Full factorial ablation included in the paper.
Statistics	Repeated runs, confidence intervals and variance reported.
Runtime	Latency, memory and throughput reported with realistic workload.
Fallback	Unknown/corrupt log behavior quantified.
Figures	All figures readable in two-column format.
Tables	All tables necessary and not overloaded.
Code/artifact	Repository or reproducibility package ready.
Co-author review	At least two co-authors have given critical positive feedback before final Prof. Kya review.

Table 18. Final checklist before sending the revised manuscript to Prof. Kya for last analysis.

19. Expected reviewer objections and how to pre-empt them

A strong revision must anticipate skeptical reviewers. The manuscript is in an area where reviewers are often suspicious of high F1-scores, heterogeneous datasets, SIEM claims and machine-learning intrusion detection. The first author should therefore write the paper as if the reviewer is asking: what is truly new, where exactly does it work, where does it fail, and can the results be reproduced?

Reviewer objection	Risk	How to answer in the paper
This is only a pipeline integration, not a scientific contribution.	Could lead to immediate rejection as an engineering report.	Define the core object as auditable family-aware specialization, then prove its value through factorial ablation and fallback analysis.
The datasets are mixed and not comparable.	Reviewers may reject aggregated conclusions.	Group datasets by family and state which claims apply to supervised, unsupervised, sequence and network-flow cases.
The routing weights are arbitrary.	The router may be seen as hand-tuned.	Add a sensitivity analysis, a calibration table and a small learned-router baseline.
HDFS performs badly.	A knowledgeable log-anomaly reviewer will notice this immediately.	Treat HDFS as evidence that family-aware routing must select sequence-aware models; add a Drain + sequence branch.
The paper compares with tools that solve different problems.	Bad comparison can weaken credibility.	Position Wazuh/OSSEC/ELK qualitatively, and compare detection only against proper detection baselines.
The high network F1 may be due to leakage or easy splits.	Cybersecurity datasets are often criticized for leakage.	Use strict split rules, duplicate checks, temporal/chunk split, class-wise reporting and error analysis.
No real SOC deployment is shown.	Operational claims can be considered speculative.	Limit claims to local analytical gateway unless a realistic ingestion experiment is added.

Table 19. Expected reviewer objections and recommended pre-emptive responses.

The safest strategy is to turn weaknesses into controlled scientific boundaries. For example, the HDFS weakness should not be hidden. It should be used to demonstrate why routing is necessary: different families require different downstream detectors. If the router can send HDFS/BGL logs to a sequence-aware branch while sending Linux/auth and network flows to tabular models, the limitation becomes evidence for the architecture.

20. Detailed quality criteria for a Class 1 version

Class 1 conference reviewers usually reward papers that combine novelty, rigor, clarity and reproducibility. The Logminer manuscript must therefore be evaluated not only as a working prototype but as a scientific argument. The following criteria should guide the next draft.

Quality criterion	Current/weak form	Target form
Research question	Can we build a tool that detects anomalies in many logs?	When and why does auditable family-aware routing improve heterogeneous log anomaly detection?
Method	A workflow with collector, parser, router and detector.	A formal architecture with explicit routing evidence, confidence, fallback and branch compatibility.
Evidence	A table of F1-scores on several datasets.	Hypothesis-driven experiments with controlled ablation, class-wise results, confidence intervals and error analysis.
Baselines	A few classical models and tool comparison.	Strong family-specific baselines, sequence baselines, monolithic baselines and calibrated fallback baselines.
Limitations	Future work list.	A regime map explaining exactly where the approach is useful and where it should not be used.
Artifact	Code exists locally.	Reproducible package with scripts, config, seeds, model cards, logs and figure regeneration.

Table 20. Difference between a prototype paper and a Class 1 conference paper.

The most important mindset change is this: the revised manuscript should not simply show that Logminer works. It should show what scientific principle Logminer demonstrates. A possible principle is: heterogeneous log anomaly detection benefits from auditable specialization when log families differ in feature contract, temporal structure and operational semantics.

21. Concrete technical strengthening of the routing mechanism

The router is the heart of the paper. At the moment it is understandable, but it can look like a manually tuned scoring table. To make it scientifically stronger, the authors should add calibration, sensitivity and failure analysis. This does not require replacing the router with a deep model. Interpretability is valuable. But the paper must prove that the rules are stable and meaningful.

- Report route confusion: for each dataset, show the predicted family, expected family and fallback rate.
- Report confidence distributions: correct route vs wrong route vs fallback route.
- Perturb each routing weight by +/-10%, +/-25% and +/-50%; show whether family decisions change.
- Add a simple learned-router baseline, for example logistic regression or random forest on schema/marker features, to show whether the hand-coded router is competitive.
- Measure unknown-family behavior separately from known-family accuracy. Unknown handling is a central operational feature, not a side detail.
- Use route explanations in the dashboard and in the audit log: route family, main evidence, confidence, fallback reason and selected detector.

Metric	Meaning	Recommended target
Route accuracy	Correct family assignments / all known-family inputs	>= 95% on known benchmark families if schema evidence is clear.

Metric	Meaning	Recommended target
Fallback precision	Unsupported inputs routed to fallback / all unsupported inputs	High, but not at the cost of rejecting too many known families.
Low-confidence rate	Inputs with confidence below threshold	Should be reported by family; high rates indicate missing schema evidence.
Route stability	Decision change under weight perturbation	Low sensitivity for well-formed known families.
Misroute severity	Impact of wrong route on detector output	Must be quantified because a wrong route can create false alerts.

Table 21. Proposed metrics for evaluating the routing mechanism as a scientific component.

22. Detailed strengthening of the correlation layer

The correlation layer is useful because analysts rarely want isolated anomaly scores. They want incident candidates. However, the current paper does not yet prove that correlation improves analyst usability. To make the correlation layer scientifically valuable, the authors should define what a good incident candidate means.

A minimal evaluation can be conducted even without a full SOC deployment. The team can sample incident candidates, ask two or three knowledgeable reviewers to label them as useful, redundant, false alarm or unclear, and report agreement. This would be more convincing than simply saying that the dashboard displays incidents.

Question	How to test it
Does correlation reduce alert volume?	Compare number of anomaly candidates before correlation and incidents after correlation.
Does correlation preserve severe cases?	Check whether high-severity candidates remain visible after grouping.
Does correlation improve interpretability?	Expert review of incident summaries, route explanations and source context.
Does correlation introduce hidden risk?	Measure whether different attack categories are merged incorrectly.
Is the time window appropriate?	Vary window size and report incident count, severity preservation and analyst usefulness.

Table 22. Correlation-layer evaluation plan.

If time is short, the paper can include a small qualitative evaluation: ten incident examples, their route evidence, model score, correlation reason and expert decision. This would immediately improve the credibility of the dashboard and correlation contribution.

23. Recommended figures for the revised paper

The current paper includes useful figures, but several are small or visually dense. The revised paper should use fewer, larger figures. Every figure should answer one scientific question. If a figure does not support a claim, it should be moved to the appendix or repository.

Recommended figure	Scientific role	Visual recommendation
Figure 1: Problem illustration	Show why heterogeneous logs break monolithic detection.	Use three log families with different fields and show the routing concept.
Figure 2: Logminer architecture	Explain the full event journey.	Use a clean left-to-right pipeline with raw log, normalized event, route, model, candidate, incident.
Figure 3: Routing confidence and fallback	Make the router auditable.	Plot confidence distributions by family and fallback cases.
Figure 4: Factorial ablation	Show what causes the performance gain.	Use grouped bars or interaction plot for router/features/model/correlation.
Figure 5: Regime map	Summarize when Logminer helps.	Axes: heterogeneity strength vs need for specialization; mark where global, family-aware and sequence branches win.
Figure 6: Operational latency	Support feasibility claim.	Show p50/p95/p99 latency, memory and

Recommended figure	Scientific role	Visual recommendation
		throughput under workload.

Table 23. Recommended figure set for a stronger paper.

24. Suggested final decision gate before submission

Before the first author submits the revised manuscript to Prof. Kya for final analysis, the team should apply the following gate. This gate protects the paper from premature submission and forces the co-authors to take responsibility for quality.

Gate	Minimum pass condition	If failed
Scientific contribution	A skeptical reader can state the novelty in one sentence.	Rewrite introduction and contribution bullets.
Ablation	At least routing, feature and model effects are separated.	Do not target Class 1 yet.
HDFS/BGL	Sequence-aware weakness is addressed experimentally or clearly bounded.	Move HDFS claims to limitation and reduce scope.
References	Related work covers all major families of prior work.	Expand before submission.
Reproducibility	All result tables can be regenerated from scripts/logs.	Freeze experiments and create artifact package.
Co-author validation	At least two co-authors have reviewed critically and approved.	Return to co-author review before Prof. Kya.

Table 24. Final internal gate before sending the paper to Prof. Kya for last review.

If these gates are met, the manuscript can be evaluated as a serious conference submission. If not, the paper should be treated as a promising internal prototype paper and developed further.

25. Quality control performed for this report

This assessment report was prepared as a professional internal review document. It was checked for consistency, readability and practical usefulness. The quality control did not claim to verify every experimental result in the original manuscript because the raw code, raw logs and full experiment repository were not provided. The control focused on the scientific assessment, the internal logic of the recommendations and the deliverable quality of this report.

Quality front	Control performed
Scientific judgment	Checked that the report distinguishes applied value, architectural novelty, experimental weakness and Class 1 readiness.
Claim discipline	Checked that recommendations avoid overclaiming and preserve the manuscript's cautious interpretation of anomaly candidates.
Actionability	Checked that the first author receives concrete experiments, tables, figures and revision tasks, not only general criticism.
Report structure	Checked that the document contains cover information, executive verdict, detailed assessment, improvement roadmap, appendices and final gate.
Tables and captions	Checked that all tables in this report have captions and clear roles.
Formatting	Checked rendered Word/PDF layout for readable headings, professional tables, page numbers and absence of clipping.
Deliverable range	Checked that the report falls within the requested 15-30 page range.
Audience alignment	Checked that the text is direct enough for the first author and useful enough for all co-authors copied on the report.

Table 25. Quality-control fronts applied to the present assessment report.

The final decision is therefore robust at the review level: Logminer is a promising manuscript with a valuable architectural idea, but it requires a major scientific strengthening cycle before it can credibly target a Class 1 international conference.

End of report